E17 metric map (exploratory): stratified − random, oriented so that > 0 = stratified looks better
colour = rank-biserial effect across tables (FMs pooled); text = median gap across tables in metric units; * Holm p < 0.05 within family
DEV (Study 7)
n = 20 tables
T1′ (Study 9)
n = 52 tables
T1′, more imbalanced half
n = 26 tables
accuracy [F1] +0.03 +0.03 -0.002 +0.000 +0.02 +0.01 +0.01 +0.005 -0.001 -0.002 -0.06 -0.05 -0.04 -0.04
top-2 accuracy (C>=3) [F1] +0.15 +0.03 +0.02 +0.000 +0.04 +0.03 +0.02 +0.02 +0.005 +0.003 +0.04 +0.01 +0.005 +0.004
weighted F1 [F1] +0.10* +0.05 +0.02 +0.001 +0.07* +0.05* +0.04* +0.02 +0.01 +0.01 +0.02 +0.01 +0.001 +0.005
Cohen's kappa [F2] +0.10* +0.06* +0.03* +0.004 +0.06* +0.05* +0.05* +0.04* +0.02* +0.01 +0.03* +0.04* +0.03* +0.04
MCC [F2] +0.10* +0.05* +0.03* +0.003 +0.06* +0.05* +0.04* +0.04* +0.02* +0.01 +0.05* +0.04* +0.03* +0.04
balanced accuracy [F3] +0.07* +0.05* +0.03* +0.01* +0.05* +0.05* +0.04* +0.04* +0.03* +0.01* +0.06* +0.06* +0.05* +0.05*
macro-F1 [F3] +0.14* +0.07* +0.03* +0.01 +0.10* +0.08* +0.06* +0.04* +0.03* +0.01* +0.08* +0.06* +0.04* +0.04
macro F0.5 [F3] +0.17* +0.07* +0.03* +0.01 +0.12* +0.09* +0.07* +0.04* +0.03* +0.01 +0.10* +0.07* +0.03* +0.03
macro F2 [F3] +0.10* +0.05* +0.03* +0.01 +0.07* +0.06* +0.05* +0.04* +0.03* +0.01* +0.06 +0.05* +0.04* +0.04
macro precision [F3] +0.19* +0.05* +0.01 -0.000 +0.13* +0.07* +0.03* +0.03* +0.003 -0.004 +0.12* +0.04* +0.003 -0.01
macro recall [F3] +0.07* +0.05* +0.03* +0.01* +0.05* +0.05* +0.04* +0.04* +0.03* +0.01* +0.06* +0.06* +0.05* +0.05*
macro Jaccard [F3] +0.09* +0.05* +0.02 +0.001 +0.05* +0.04* +0.04* +0.03* +0.02* +0.01 +0.03 +0.03 +0.02 +0.01
AUC OvR macro [F4] +0.10* +0.02* +0.01 +0.002 +0.06* +0.04* +0.02* +0.01* +0.002 +0.000 +0.06* +0.05* +0.01 +0.01
AUC OvR weighted [F4] +0.09* +0.02* +0.01 +0.002 +0.05* +0.03* +0.02* +0.01* +0.002 +0.000 +0.03* +0.02* +0.004 +0.01
AUC OvO macro [F4] +0.10* +0.03* +0.01 +0.002 +0.06* +0.04* +0.02* +0.01* +0.002 +0.001 +0.07* +0.06* +0.01 +0.01
AP macro [F4] +0.10* +0.03* +0.02 +0.000 +0.04* +0.03* +0.02* +0.01* +0.003 +0.000 +0.04* +0.04* +0.02 +0.02
AUC OvR macro [F5] -0.003 +0.01 +0.01 +0.001 -0.01 -0.001 +0.01 +0.01 +0.002 +0.000 -0.01 +0.004 +0.01 +0.01
AUC OvR weighted [F5] -0.003 +0.01 +0.01 +0.001 -0.02 -0.001 +0.004 +0.01 +0.001 +0.000 -0.03 +0.000 +0.004 +0.01
AUC OvO macro [F5] -0.01 +0.01 +0.01 +0.001 -0.02 -0.002 +0.004 +0.01 +0.002 +0.001 -0.03 +0.000 +0.01 +0.01
AP macro [F5] -0.003 +0.01 +0.01 -0.001 -0.02 -0.001 +0.005 +0.01 +0.002 +0.000 -0.04 -0.001 +0.01 +0.02
log loss [F6] +3.27* +0.30* +0.06 -0.003 +2.27* +0.73* +0.18* +0.04 +0.000 -0.01 +1.85* +0.39* -0.01 -0.04
Brier [F6] +0.16* +0.04 +0.02 +0.000 +0.13* +0.03 +0.01 +0.005 -0.004 -0.005 +0.04 -0.03 -0.04 -0.04
D2 log loss [F6] +3.66* +0.37* +0.05 -0.003 +2.70* +0.82* +0.19* +0.04 -0.001 -0.01 +2.55* +0.45* -0.02 -0.07
D2 Brier [F6] +0.35* +0.06 +0.02 -0.000 +0.21* +0.04 +0.02 +0.003 -0.01 -0.01 +0.06 -0.06 -0.11* -0.11*
ECE (top-label) [F7] +0.11* +0.02 +0.01 +0.003 +0.10* +0.02* +0.01 +0.003 +0.002 -0.004 +0.05 +0.003 -0.003 -0.01
classwise ECE [F7] +0.09 +0.01 +0.005 +0.001 +0.09* +0.01 +0.01 -0.001 -0.000 -0.003 +0.01 -0.03* -0.05* -0.07*
CITL (L1) [F7] +0.44* +0.07 +0.02 +0.003 +0.27* +0.07 +0.05 +0.02 -0.02 -0.01 +0.09 -0.09* -0.21* -0.19*
KL(test prior || mean p) [F7] +3.30* +0.35* +0.02 +0.000 +2.38* +0.73* +0.20* +0.03* +0.002 +0.000 +1.79* +0.43* -0.02 -0.04*
log LR+ [F8] +0.12 -0.05 -0.12 -0.12 +0.01 -0.15* -0.09* -0.15* -0.28* -0.28* -0.35* -0.33* -0.54* -0.67*
log LR- [F8] +0.43* +0.19 +0.34 +0.10 +0.20* +0.18* +0.11* +0.17* +0.23* +0.22* +0.23* +0.21* +0.36* +0.40*
1C 3C 5C 10C 1C 2C 3C 5C 10C 20C 1C 3C 10C 20C
rank-biserial (−1: random better, +1: stratified better)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00